# ForgeNet-X

AI Handwriting Imitation & Signature Forgery Analysis System

## Visual Confirmation Report

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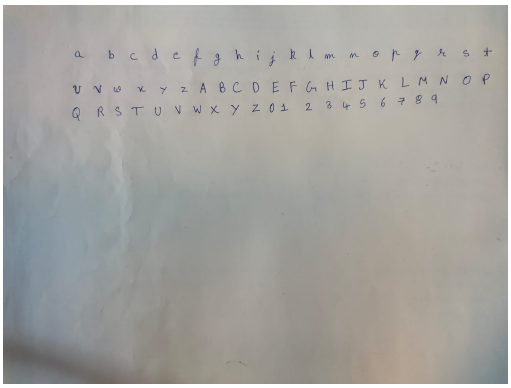
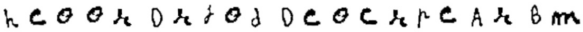
| | |
|---|---|
| **System** | ForgeNet-X v1.0 |
| **Purpose** | Handwriting imitation & signature forgery detection |
| **Method** | Image processing + Computer vision (OpenCV · scikit-image) |
| **Libraries** | Flask · OpenCV · NumPy · scikit-image · ReportLab |
| **Report** | Full pipeline visuals, segmentation atlas, forgery metrics |

## 1. Handwriting Analysis

The original handwriting sample uploaded by the user (left) and the synthetically generated text rendered in the same writing style (right).

| Original Handwriting Sample | Generated Handwriting Output |
| --- | --- |
|  |  |

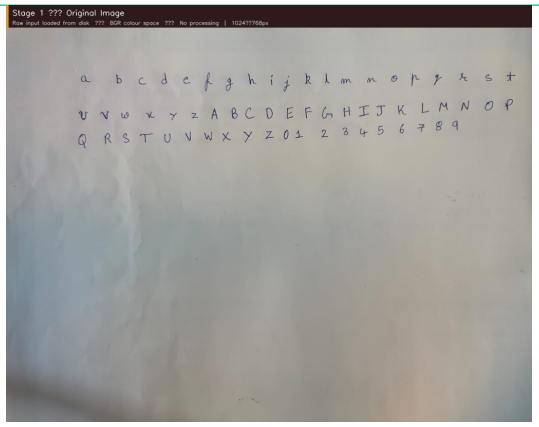
## 2. Image Preprocessing Pipeline

Every uploaded image passes through five sequential processing stages before characters are extracted. Each stage is shown below at full report width. The coloured banner on each image was added by ForgeNet-X to identify the stage name, description, and key pixel statistics.

**Stage 1**

**BGR colour image loaded directly from disk.**

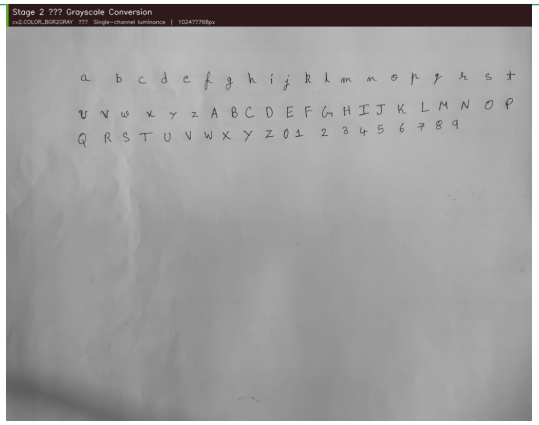
No processing applied — serves as the ground-truth reference.




**Stage 2**

**Converted to single-channel luminance using cv2.COLOR_BGR2GRAY.**

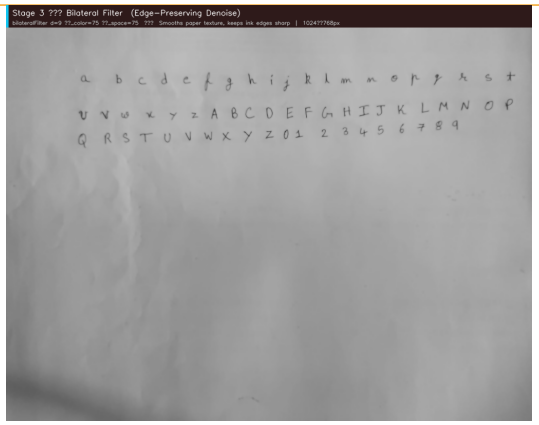
Eliminates colour variation so binarisation operates on intensity only.




**Stage 3**

**Bilateral filter (d=9, σ_colour=75, σ_space=75) — edge-preserving denoise.**

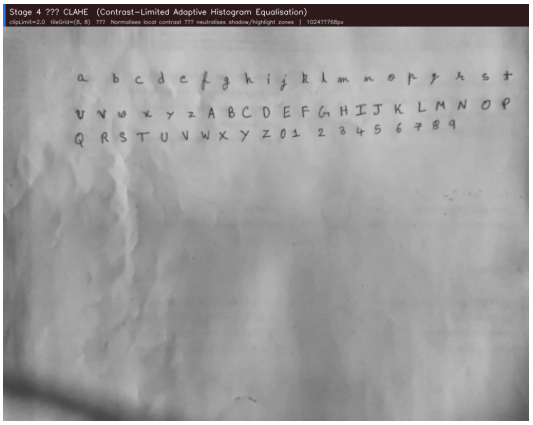
Smooths paper grain and texture while keeping ink-stroke edges sharp.




**Stage 4**

**CLAHE (Contrast-Limited Adaptive Histogram Equalisation) on bilateral image.**

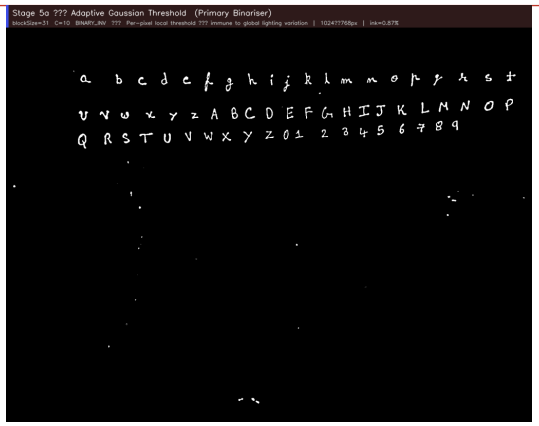
Normalises local contrast tile-by-tile — neutralises shadows and highlights.




**Stage 5**

**Adaptive Gaussian Threshold — primary binariser (blockSize=31, C=10).**

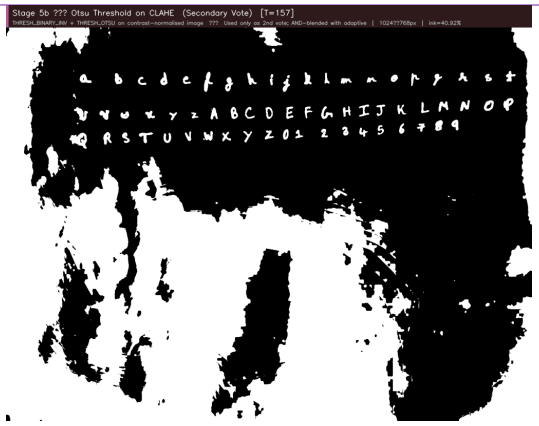
Each pixel gets its own local threshold — immune to global lighting variation.




**Stage 6**

**Otsu threshold applied to the CLAHE image — used as a secondary vote.**

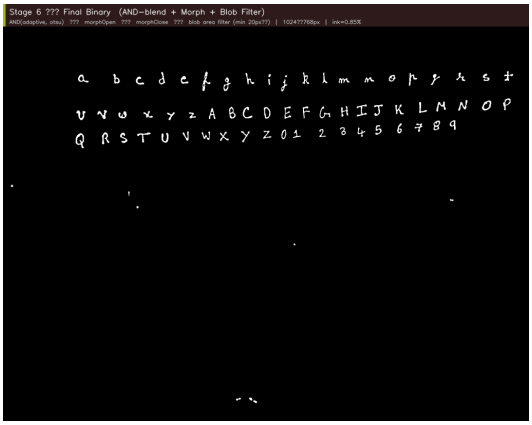
AND-blended with adaptive: only pixels both methods call ink are kept.




**Stage 7**

**AND(adaptive, otsu) → morphOpen → morphClose → blob area filter (≥20px²).**

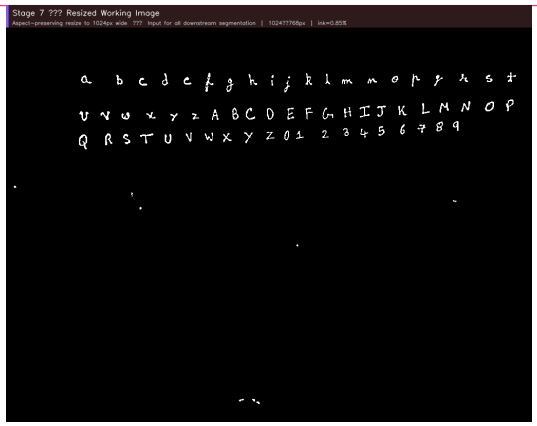
Final cleaned binary: false-positive paper noise and tiny specks removed.




**Stage 8**

**Aspect-preserving resize to 1024px wide + re-binarise after interpolation.**

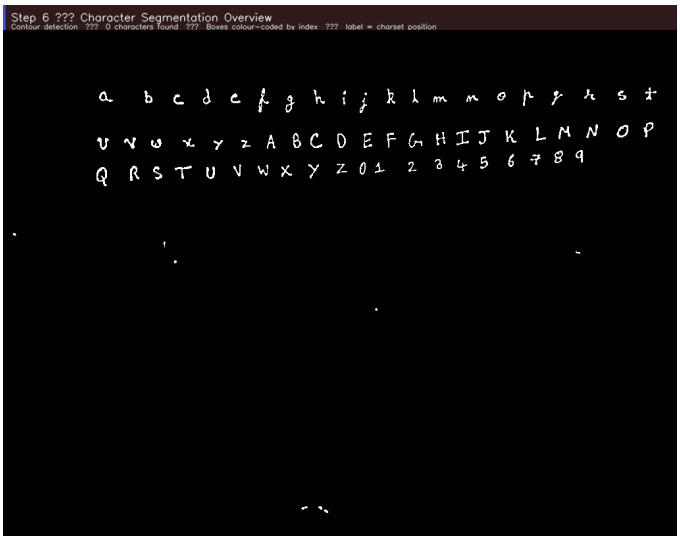
Standardised working image used for all character segmentation.



## 3. Character Segmentation

### 3a — Segmentation Overview (Bounding Boxes)

The preprocessed binary image is scanned for external contours. Each contour that passes the size filter (width 5–200 px, height ≥ 8 px) is wrapped in a coloured bounding box. The tag above each box shows **#index / charset-label**, where the label is the character that would be assigned to that crop if the sheet follows a standard a–z practice layout.




### 3b — Character Atlas Sheet (Individual Crops)

Every extracted character crop is shown in the grid below. Each tile displays the ink pixels in the colour assigned to that crop's index. The dark strip beneath each tile shows the crop's **#index** (top-left) and its **assigned label** (centre). Labels are heuristic — they follow the left-to-right reading order of the source sheet mapped onto the charset: a–z → A–Z → 0–9 → punctuation.

### 3c — Charset Label Assignment Map

The table below shows exactly which label is assigned to each crop index. Index 0 → 'a', index 1 → 'b', … index 25 → 'z', index 26 → 'A', etc. If more crops are found than charset entries, they are marked '?'.

| # | Label | # | Label | # | Label | # | Label | # | Label | # | Label | # | Label | # | Label | # | Label | # | Label | # | Label | # | Label | # | Label |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| 0 | "a" | 1 | "b" | 2 | "c" | 3 | "d" | 4 | "e" | 5 | "f" | 6 | "g" | 7 | "h" | 8 | "i" | 9 | "j" | 10 | "k" | 11 | "l" | 12 | "m" |
| 13 | "n" | 14 | "o" | 15 | "p" | 16 | "q" | 17 | "r" | 18 | "s" | 19 | "t" | 20 | "u" | 21 | "v" | 22 | "w" | 23 | "x" | 24 | "y" | 25 | "z" |
| 26 | "A" | 27 | "B" | 28 | "C" | 29 | "D" | 30 | "E" | 31 | "F" | 32 | "G" | 33 | "H" | 34 | "I" | 35 | "J" | 36 | "K" | 37 | "L" | 38 | "M" |
| 39 | "N" | 40 | "O" | 41 | "P" | 42 | "Q" | 43 | "R" | 44 | "S" | 45 | "T" | 46 | "U" | 47 | "V" | 48 | "W" | 49 | "X" | 50 | "Y" | 51 | "Z" |
| 52 | "0" | 53 | "1" | 54 | "2" | 55 | "3" | 56 | "4" | 57 | "5" | 58 | "6" | 59 | "7" | 60 | "8" | 61 | "9" | 62 | "." | 63 | "," | 64 | "!" |
| 65 | "?" | 66 | ";" | 67 | ":" | 68 | "'" | 69 | "'" | 70 | "(" | 71 | ")" | 72 | "-" | | | | | | | | | | |

## 5. Conclusion

No analysis data available.

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